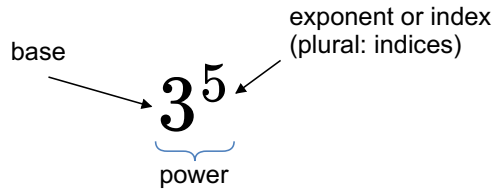
Algebra and Functions

Chapter 1 · Original slide 4

1: Index Laws

4



$a^m \times a^n = a^{m+n}$
$a^m \div a^n = a^{m-n}$
$(a^m)^n = a^{mn}$
$(ab)^n = a^n b^n$

Some also refer to the 5 as the ‘power’. But this would lead to unfortunate ambiguous phrases like “when we multiply two powers together, we add the powers” instead of “when we multiply two powers together, we add the indices”.

Simplify $(a^3)^2 \times 2a^2$

Simplify $(4x^3y)^3$

Simplify $2x^2(3+5x) - x(4-x^2)$

Simplify $\frac{x^3-2x}{3x^2}$

Chapter 1 · Original slide 5

Test Your Understanding

5

1 Simplify $\left(\frac{2a^5}{a^2}\right)^2 \times 3a$

2 Simplify $\frac{2x+x^5}{4x^3}$

3 Expand and simplify
$2x(3-x^2) - 4x^3(3-x)$

4 Simplify $2^x \times 3^x$

Algebra and Functions

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Exercise 1A – Pearson Pure Y1 page 4

6

2 Expand and simplify if possible:

o $x(3x^2 - 2x + 5)$
p $7y^2(2 - 5y + 3y^2)$
q $-2y^2(5 - 7y + 3y^2)$
r $7(x - 2) + 3(x + 4) - 6(x - 2)$
s $5x - 3(4 - 2x) + 6$
t $3x^2 - x(3 - 4x) + 7$
u $4x(x + 3) - 2x(3x - 7)$
v $3x^2(2x + 1) - 5x^2(3x - 4)$

3 Simplify these fractions:

d $\frac{8x^3+5x}{2x}$ **e** $\frac{7x^7+5x^2}{5x}$ **f** $\frac{9x^5-5x^3}{3x}$

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Exercise 1A – Pearson Pure Y1 page 4

7

Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 [MAT 2006 1A] Which of the following numbers is the largest?

$((2^3)^2)^3, \quad (2^3)^{(2^3)}, \quad 2^{((3^2)^3)}, \quad 2^{(3^{(2^3)})}$

2 [MAT 2006 1B] If $N = 2^k \times 4^m \times 8^n$ where k, m, n are positive whole numbers. Then N will definitely be a square number whenever:

k is even;
$k + n$ is odd;
k is odd but $m + n$ is even;
$k + n$ is even.

Algebra and Functions

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4: Negative and Fractional Indices

17

$$a^0 = 1$$
$$a^{\frac{1}{m}} = \sqrt[m]{a}$$
$$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$$
$$a^{-m} = \frac{1}{a^m}$$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3.
Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $27^{-\frac{1}{3}}$

Evaluate $32^{\frac{2}{5}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

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Writing a surd using indices

18

If $9\sqrt{3} = 3^k$, determine the value of k.

Algebra and Functions

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Further Examples

19

[Edexcel IGCSE May14-4H]
Given that

$$\left(2^{\frac{1}{2}}\right)^{n} = \frac{2^{x}}{8^{y}}$$

Express n in terms of x and y.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^{x}$, express the following in terms of y.

$$\frac{1}{4^{2x-3}}$$

Write your expression in its simplest form.

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Test Your Understanding

20

Edexcel Paper 2 – May 2019

Given $2^{x} \times 4^{y} = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

Algebra and Functions

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Exercise 1D

21

4 a Find the value of $81^{\frac{1}{4}}$. **b** Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants. **a** $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge

[MAT 2007 1A] Let r and s be integers. Then $\frac{6^{r+s}\times 12^{r-s}}{8^r \times 9^{r+2s}}$ is an integer if $r+s \leq 0$; $s \leq 0$; $r \leq 0$; $r \geq s$.

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5: Surds

22

Recap:

A surd is a root of a number that does not simplify to a rational number.

Laws:

$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$

Note: A *rational* number is any which can be expressed as $\frac{a}{b}$ where a, b are integers. $\frac{2}{3}$ and $\frac{4}{1} = 4$ are rational numbers, but π and $\sqrt{2}$ are not.

$\sqrt{3} \times 2$

$3\sqrt{5} \times 2\sqrt{5}$

$\sqrt{8} = \sqrt{4}\sqrt{2}$

$\sqrt{12} + \sqrt{27}$

$\left(\sqrt{8}+1\right)\left(\sqrt{2}-3\right)$

Algebra and Functions

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Exercise 1E – Pure Y1 Page 13

23

3 Simplify $\sqrt{75} - \sqrt{12}$ giving your answer in the form $a\sqrt{3}$, where a is an integer.

Extension

1 [SMC 2014 Q24] Which of the following is smallest?
$10 - 3\sqrt{11}; \quad 8 - 3\sqrt{7}; \quad 5 - 2\sqrt{6}; \quad 9 - 4\sqrt{5}; \quad 7 - 4\sqrt{3}.$

2 [SMC 2012 Q21] Which of the following numbers does not have a square root in the form $x + y\sqrt{2}$, where x and y are positive integers?
$17 + 12\sqrt{2}; \quad 22 + 12\sqrt{2}; \quad 38 + 12\sqrt{2}; \quad 54 + 12\sqrt{2}; \quad 73 + 12\sqrt{2}.$

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6: Rationalising The Denominator

24

Here's a surd. What could we multiply it by such that it's no longer an irrational number?

$$\sqrt{5} \times \square = \square$$

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Examples

25

$\frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$

$\frac{6}{\sqrt{3}} = 2\sqrt{3}$

$\frac{7}{\sqrt{7}} = \frac{7\sqrt{7}}{7} = \sqrt{7}$

$\frac{15}{\sqrt{5}} + \sqrt{5} = 4\sqrt{5}$

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More Complex Denominators

26

You've seen 'rationalising a denominator', the idea being that we don't like to divide things by an irrational number.

But what do we multiply the top and bottom by if we have a more complicated denominator?

$\frac{1}{\sqrt{2}+1} \times \quad = \quad$

Algebra and Functions

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More Examples

27

$\frac{3}{\sqrt{6}-2} \times$ ☐ $=$ ☐

$\frac{4}{\sqrt{3}+1} \times$ ☐ $=$ ☐ 2 ☐

$\frac{3\sqrt{2}+4}{5\sqrt{2}-7} \times \frac{5\sqrt{2}+}{5\sqrt{2}+}$ ☐ ☐

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Test Your Understanding

28

Rationalise the denominator and simplify $\frac{4}{\sqrt{5}-2}$ ☐

Rationalise the denominator and simplify $\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$

$\frac{2\sqrt{3}}{3\sqrt{3}}$ ☐

$= \frac{1}{}$

$= \frac{1}{}$

AQA IGCSE FM June 2013 Paper 1

Solve $y\left(\sqrt{3}-1\right)=8$

Give your answer in the form $a+b\sqrt{3}$ where a and b are integers.

$y=$ ☐

Algebra and Functions

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Writing surd expressions in power form

29

This is not in the textbook, but a common type of question in exams, often asked in conjunction with differentiation (a later chapter).

Express $\frac{(2+\sqrt{x})^2}{x}$ as powers of x.

Test Your Understanding

Express $\frac{(x+\sqrt{x})^2}{2\sqrt{x}}$ as powers of x.

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Exercise 1F (Page 15)

30

or alternatively: (not in textbook)

1 Rationalise the denominator and simplify the following:

a $\frac{1}{\sqrt{5}+2} =$

b $\frac{\sqrt{3}}{\sqrt{3}-1} = \frac{3+\sqrt{3}}{2}$

c $\frac{\sqrt{5}+1}{\sqrt{5}-2} = 7+3\sqrt{}$

d $\frac{2\sqrt{3}-1}{3\sqrt{3}+4} = 2-\sqrt{}$

e $\frac{5\sqrt{5}-2}{2\sqrt{5}-3} = 4+\sqrt{}$

2 Expand and simplify:
$\left(\sqrt{5}+3\right)\left(\sqrt{5}-2\right)\left(\sqrt{5}+1\right)$

3 Rationalise the denominator, giving your answer in the form $a+b\sqrt{3}$.
$\frac{3\sqrt{3}+7}{3\sqrt{3}-5} =$

4 Solve $x\left(4-\sqrt{6}\right)=10$ giving your answer in the form $a+b\sqrt{6}$.
$x = \frac{10}{4-\sqrt{6}} = 4+\sqrt{6}$

Solv

5 $y = \frac{3+\sqrt{2}}{1+\sqrt{2}} = 2\sqrt{2}-1$
Simplify:

6

Algebra and Functions $\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}} = 2a+1-2\sqrt{a}\sqrt{a+1}$

Chapter 2 · Original slide 5

1: Solving Quadratic Equations

5

$$x^2 + 5x = 6$$

There are three ways of solving a quadratic equation.
One is by *completing the square*, which we'll do later.
What are the other two ways?

By factorisation

Using the quadratic formula.

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1: Solving Quadratic Equations

6

Solving without factorising

If the subject only appears once however, it might be easier not to expand out/factorise:

$$(x-1)^2 = 5$$

Algebra and Functions

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Test Your Understanding

7

1 Solve $(x+3)^2 = x+5$ using factorisation.

2 Solve $(2x+1)^2 = 5$

3 Solve $\sqrt{x+3} = x-3$

4 Solve $2x + \sqrt{x} - 1 = 0$

"Use of Technology":
The 'equation' mode on your calculator will solve quadratics in the form $ax^2+bx+c=0$. When you're asked for the 'order', use 2 (we'll see why later).

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Exercise 2A/2B Pure Y1 Pages 20-22

8

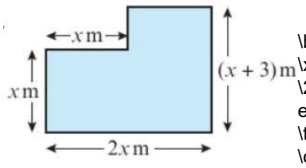
3 Solve the following equations:

i $6x^2 - 7 = 11x$

h $5x^2 - 10x^2 = -7 + x + x^2$

\
x
\
x

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.



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5 Solve the equation $5x + 3 = \sqrt{3x+7}$.

Algebra and Functions

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Exercise 2A/2B Pure Y1 Pages 20-22

9

2 Solve the following equations using the quadratic formula. Give your answers to three significant figures.

h $2x^2 + 11x = 5x^2 - 18$

3 For each of the equations below, choose a suitable method and find all of the solutions. Where necessary, give your answers to three significant figures.

e $(2x + 8)^2 = 100$ **f** $6x^2 + 6 = 12x$

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Exercise 2A/2B Pure Y1 Pages 20-22

10

Extension: (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 (i) Use the substitution $\sqrt{x} = y$ $(y \geq 0)$ to find the real root of the equation $x + 3\sqrt{x} - \frac{1}{2} = 0$.

(ii) Find all real roots of the following equations:

(a) $x + 10\sqrt{x + 2} - 22 = 0$;

(b) $x^2 - 4x + \sqrt{2x^2 - 8x - 3} - 9 = 0$.

Algebra and Functions

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2: Completing the Square

11

"Completing the square" means putting a quadratic in the form $(x+a)^2+b$ or $a(x+b)^2+c$

The **underlying reason we do this is because x only appears once in the expression** (e.g. in $(x+2)^2+3$ vs x^2+4x+7), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

If we have a completed square:

$$(x+4)^2-7=0$$

we saw at the start of the chapter how we could rearrange to make x the subject.
Indeed using the quadratic formula is actually solving the quadratic by completing the square – it's just someone has done the work for us already!

b. Sketching Quadratics

We'll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

c. In integration

In Further Maths, completing the square allows us to 'integrate' expressions like:

$$\int \frac{1}{x^2-4x+5}\,dx$$

(you will cover integration later this module)

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Completing the Square Recap

12

Expand:

$(x+\mathbf{9})^2 = x^2+$

$(x-\mathbf{5})^2 = x^2-$

What do you notice about the relationship between the bold numbers?

Algebra and Functions

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Completing the Square

13

Express $2x^2 + 12x + 7$ in the form $a(x+b)^2 + c$

$= 2\left(x\right.$

$= 2\left(\left(\right.$

$= 2(x$

$= 2(x$

Factorise out coefficient of x^2. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x+c)^2$

$= -3x^2$

$= -3\left(x\right.$

$= -3\left(\left(\right.$

$= -3(x$

$= 8 - 3($

It may help to write in the form $ax^2 + bx + c$ first.

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Test Your Understanding

14

Express $3x^2 - 18x + 4$ in the form $a(x+b)^2 + c$

$= 3\left(\right.$

$= 3\left(\right.$

$= 3(x$

$= 3(x$

Express $20x - 5x^2 + 3$ in the form $a - b(x+c)^2$

$= -$

$= -$

$= -$

$= -$

$= 23$

Algebra and Functions

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Solving by Completing the Square

15

Solve the equation:

$$3x^2 - 18x + 4 = 0$$

x^2

$(x$

$(x$

x

x

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Exercise 2C/2D Pure Y1 Page 23-24

16

1 Complete the square for the expressions: **a** $x^2 + 4x$ **b** $x^2 - 6x$ **c** $x^2 - 16x$ **d** $x^2 + x$ **e** $x^2 - 14$

2 Complete the square for the expressions: **a** $2x^2 + 16x$ **b** $3x^2 - 24x$ **c** $5x^2 + 20x$ **d** $2x^2 - 5x$ **e** $8x - 2x^2$

Algebra and Functions

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Exercise 2C/2D Pure Y1 Page 23-24

17

4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants a and b

5 Write $2 + 0.8\mathrm{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A \mathrm{and} B$ are constants to be determined

4 By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

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Proving the Quadratic Formula

18

If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$ax^2 + bx + c = 0$

$x^2 + \frac{b}{a}$

$\left(x + \frac{b}{2}\right.$

$\left(x + \frac{b}{2}\right.$

$\left(x + \frac{b}{2}\right.$

$x + \frac{b}{2a}$

$x = \frac{-b}{}$

Just use exactly the same method as you usually would!

Algebra and Functions

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Test Your Understanding

24

1 Find the minimum value of $f(x) = 2x^2 + 12x - 5$ and state the value of x for which this occurs.

2 Find the roots of the function $f(x) = 2x^2 + 3x + 1$

3 Find the roots of the function $f(x) = x^4 - x^2 - 6$

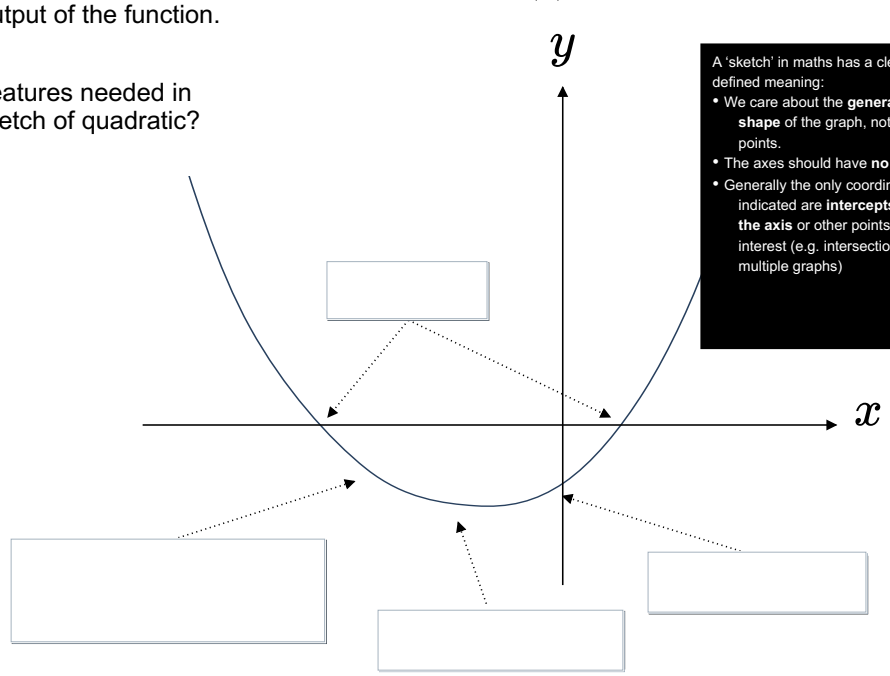
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4: Quadratic Graphs

26

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

Features needed in sketch of quadratic?



A 'sketch' in maths has a clearly defined meaning:
- We care about the **general shape** of the graph, not exact points.
- The axes should have **no scale**.
- Generally the only coordinates indicated are **intercepts with the axis** or other points of interest (e.g. intersections of multiple graphs)

Algebra and Functions

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Example

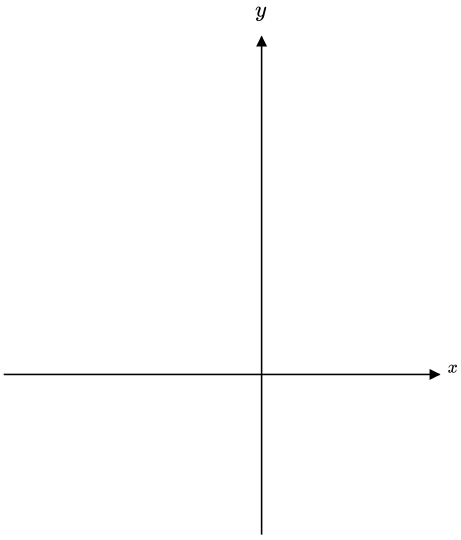
27

Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

y-intercept:

Turning point:



Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

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Example

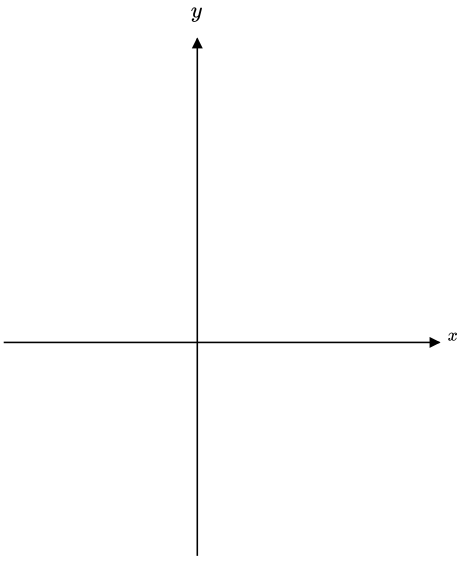
28

Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

Roots:

y-intercept:

Turning point:



Line of reflection:

Algebra and Functions

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Test Your Understanding

29

Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

a $y = x^2 + 4$

c $y = 5x + 3 - 2x^2$

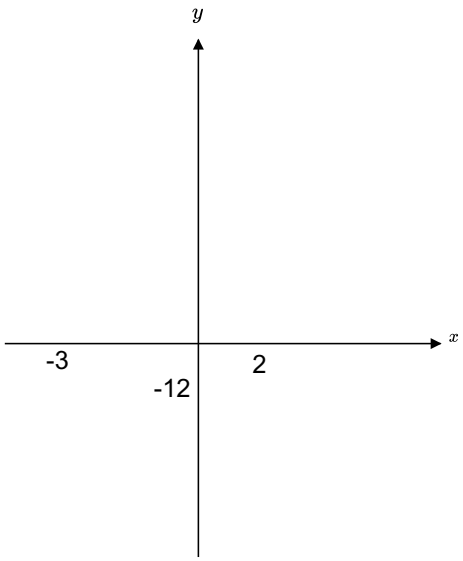
b $y = x^2 - 7x + 10$

d $y = x^2 + 4x + 11$

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Determining the Equation using a Graph

30



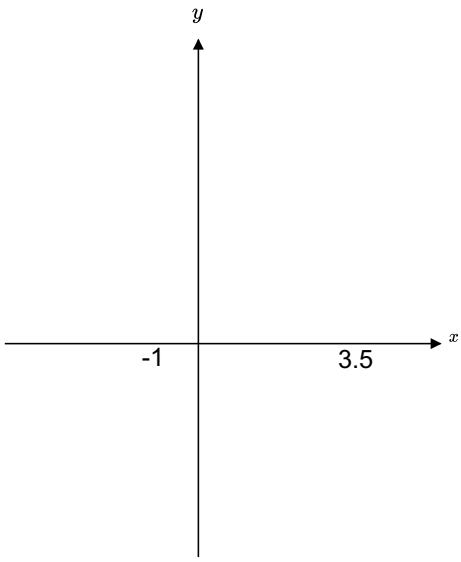
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

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Further Example

31

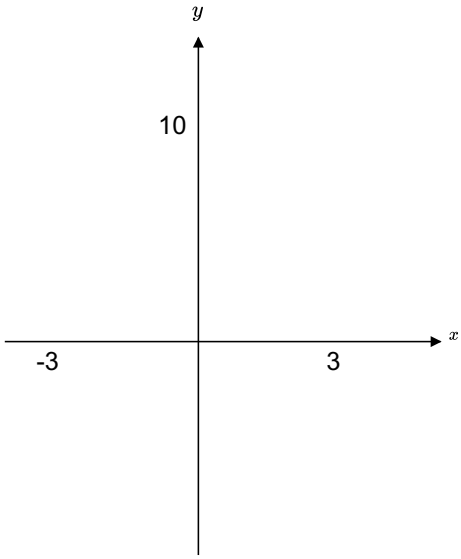


Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

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Test Your Understanding

32



Determine an equation of this quadratic graph.

Algebra and Functions

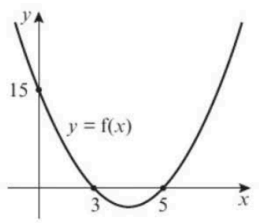
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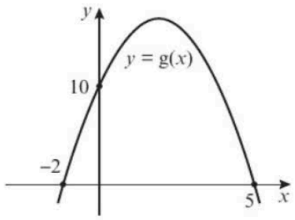
33

2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a, b, c for each function.

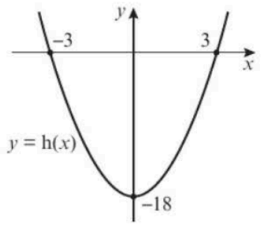
a



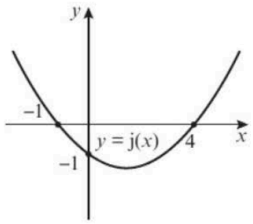
b



c



d



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Exercise 2F Pure Y1 Page 30

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3 The graph of $y = ax^2 + bx + c$ has a minimum at $(5, -3)$ and passes through $(4, 0)$. Find the values of a, b, c.

Extension Question:

[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?

$y = x^2$

$y = x^2 - 2x$

$y = x^2 + 2x + 2$

Algebra and Functions

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Starter

35

How many **distinct** real solutions do each of the following have?

$x^2 - 12x + 36 = 0$

$x^2 + x + 3 = 0$

$x^2 - 2x - 1 = 0$

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5: The Discriminant

36

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note: Roots of a **function** f are the values of x such that $f(x) = 0$.
Similarly the roots of an **equation** are solutions to an equation in the form $f(x) = 0$

Looking at this formula, when in general do you think we have:
- No real roots?
- Equal roots?
- Two distinct roots?

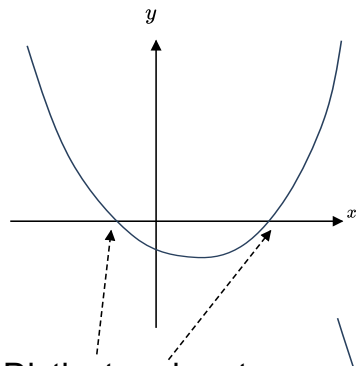


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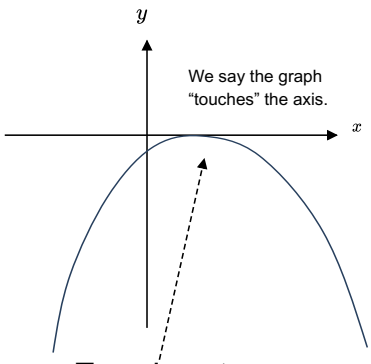
5: The Discriminant

37



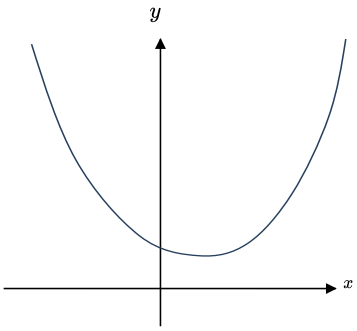
Distinct real roots

$\mathbf{b^2-4ac>0}$



Equal roots

$\mathbf{b^2-4ac=0}$



No real roots

$\mathbf{b^2-4ac<0}$

Chapter 2 · Original slide 38

Just for your interest…

38

Why do we say “Equal Roots” not “One root”?

$$x^2 - 12x + 36 = 0$$

Using the quadratic formula gives us the same value in both $+$ and $-$ cases: $x = 6$.

You might wonder why we say “*it has one repeated root*” or “*it has equal roots*”, i.e. indicating we have **2 roots** (but with the same value). Why not say it has 1 root?

It is due to the Fundamental Theorem of Algebra:

“Every polynomial of order n has exactly n roots.”

A **polynomial** is an expression with non-negative integer powers of x, i.e. $a + bx + cx^2 + dx^3 + \ldots$ All linear, quadratic and cubic expressions are examples of polynomials.

The **order** of a polynomial is its highest power of x. So the order of a quadratic is 2, and a cubic 3.

These roots might be repeated or might not be ‘real’ roots. $\sqrt{-1}$ is known as a **complex number**, which you will encounter if you do FM. But **it is still a value**!

The theorem means that a quadratic (order 2) will **always** have 2 roots. This is why you should say “no **real** roots” when $b^2 - 4ac < 0$ rather than “no roots”, because there are still roots – it’s just they’re not ‘real’! Similarly we must say “*equal roots*” because there are still 2 roots.

Despite the theorem being a simple statement, it was only until 1806 that it was first proven by **Argand**. Clearly by using the quadratic formula we can show a quadratic equation has 2 roots. We can use similar formulas to show that a cubic has 3 roots and a quartic 4 roots. But there is **provably** no such formulae for order 5 (quintics) and beyond. So we have to prove for example that 5 roots exist for a quintic, despite us having no way to find these exact roots!

One side result of the Fundamental Theorem of Algebra is that every polynomial can be written as a product of linear and/or quadratic expressions.

Leibniz claimed in 1702 that a polynomial of the form $x^4 + a^4$ cannot be written in this way. He then got completely burned by Euler in 1742 who managed to do so:

$$x^4 + a^4 = \left(x^2 + a\sqrt{2}x + a^2\right)\left(x^2 - a\sqrt{2}x + a^2\right)$$

There are various other ‘Fundamental Laws’. The ‘**FL of Arithmetic**’ you encountered at KS3, which states that “*every positive integer* > 1 *can be written as a product of primes in one way only*”. You will encounter the ‘**FL of Calculus**’ in Chapter 13.

Algebra and Functions

Chapter 2 · Original slide 39

Quickfire Questions

39

Equation	Discriminant	Number of Distinct Real Roots
$x^2 + 3x + 4 = 0$		
$x^2 - 4x + 1 = 0$		
$x^2 - 4x + 4 = 0$		
$2x^2 - 6x - 3 = 0$		
$x - 4 - 3x^2 = 0$		
$1 - x^2 = 0$		

Chapter 2 · Original slide 40

Problems involving the discriminant

40

8. The equation $x^2 + 2px + (3p + 4) = 0$, where p is a positive constant, has equal roots.

(a) Find the value of p. (4)

(b) For this value of p, solve the equation $x^2 + 2px + (3p + 4) = 0$. (2)

a) $a =$ $b =$ $c =$ 4

$(2p)^2 - 4($

$4p^2 - 12p$

$p^2 - 3p -$

$(p + 1)(p$

$p = 4$

b)

$(x$

Tip: Always start by writing out a, b and c explicitly.

Algebra and Functions

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Exercise 2G Pure Y1 Page 32

41

2 Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions.

3 Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution.

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Exercise 2G Pure Y1 Page 32

42

4 Given that the function $\mathrm{f}(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s.

5 Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions.

\

Algebra and Functions

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Exercise 2G Pure Y1 Page 32

43

6 The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.
a Find the value of p.
b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$.

7 $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.
a Find the discriminant of $h(x)$ in terms of k.
b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k.

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Test Your Understanding

44

$$x^2 + 5kx + (10k + 5) = 0$$

where k is a constant.
Given that this equation has equal roots, determine the value of k.

Find the range of values of k for which $x^2 + 6x + k = 0$ has two distinct real solutions.

Algebra and Functions

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Exercise 2G Pure Y1 Page 32

45

Extension Questions:

1 [MAT 2009 1C] Given a real constant c, the equation

$$x^4 = (x-c)^2$$

Has four real solutions (including possible repeated roots) for:
A) $c \leq \frac{1}{4}$
B) $-\frac{1}{4} \leq c \leq \frac{1}{4}$
C) $c \leq -\frac{1}{4}$
D) all values of c

Square rooting: $\mathbf{x^2 = \pm(x-c)}$

2 [MAT 2006 1B] The equation $\left(2 + x - x^2\right)^2 = 16$ has how many real root(s)?
A)

3 [MAT 2011 1B] A rectangle has perimeter P and area A. The values P and A must satisfy:
A) $P^3 > A$
B) $A^2 > 2P + 1$
C) $P^2 \geq 16A$
D) $PA > A + P$
E)

Let x and y be the width and height.
A = xy, P = 2x + 2y

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Modelling

46

The new A Level has a particular emphasis of the application of theory to real-life situations. A mathematical model is the maths used to model such a situation, possibly with some simplifying assumptions.

Example (from textbook): A spear is thrown over level ground from the top of a tower. The height, in metres, of the spear above the ground after t seconds is modelled by the function: $h(t) = 12.25 + 14.7t - 4.9t^2, \quad t \geq 0$

a) Interpret the meaning of the constant term 12.25 in the model.
b) After how many seconds does the spear hit the ground?
c) Write $h(t)$ in the form $A - B(t-C)^2$, where A, B and C are constants to be found.
d) Using your answer to part c or otherwise, find the maximum height of the spear above the ground, and the time at which this maximum height is reached?

a

$h(t) = 0 \qquad 12.25 + 14.7t - 4.9t^2$

b

The maximum height of the spear is 23.275m, 1.5 seconds after spear is thrown.

c

d

Algebra and Functions

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Exercise 2H Pure Y1 Page 34

47

2 A car manufacturer uses a model to predict the fuel consumption, y miles per gallon (mpg), for a specific model of car travelling at a speed of x mph. $y = -4.81x^2 + 0.975x + 16$, $x > 0$.

a Use the model to find two speeds at which the car has a fuel consumption of 32.5 mpg.
b Rewrite y in the form $A - B(x - C)^2$, where A, B, C are constants to be found.
c Using your answer to part b, find the speed at which the car has the greatest fuel efficiency.
d Use the model to calculate the fuel consumption of a car travelling at 120 mph.
Comment on the validity of using this model for very high speeds.

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Exercise 2H Pure Y1 Page 34

48

3 A fertiliser company uses a model to determine how the amount of fertiliser used, f kilograms per hectare, affects the grain yield y, measured in tonnes per hectare. $y = 4 + 0.3f - 0.0006f^2$.

a According to the model, how much grain would each hectare yield without any fertiliser? (1 mark)
b One farmer currently uses 20 kilograms of fertiliser per hectare. How much more fertiliser would he need to use to increase his grain yield by 1 tonne per hectare? (4 marks)

Algebra and Functions

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Exercise 2H Pure Y1 Page 34

49

4 A football stadium has 25000 seats. The football club know from past experience that they will sell only 10000 tickets if each ticket costs $\pounds 30$. They also expect to sell 1000 more tickets every time the price goes down by $\pounds 1$. **a** The number of tickets sold t can be modelled by the linear equation $t = M - 1000p$, where $\pounds p$ is the price of each ticket and M is a constant. Find the value of M. (1 mark)

The total revenue, £r, can be calculated by multiplying the number of tickets sold by the price of each ticket. This can be written as $r = p(M - 1000p)$. **b** Rearrange r into the form $A - B(p - C)^2$, where A, B, C are constants to be found. (3 marks) **c** Using your answer to part b or otherwise, work out how much the football club should charge for each ticket if they want to make the maximum amount of money. (2 marks)

Chapter 3 · Original slide 3

Solutions sets

3

The solution(s) to an equation may be:

A single value:	$2x + 1 = 5$
Multiple values:	$x^2 + 3x + 2 = 0$
An infinitely large set of values:	$x > 3$
No (real) values!	$x^2 = -1$
Every value!	$x^2 + x = x(x + 1)$

The point is that you shouldn't think of the solution to an equation/inequality as an 'answer', but a **set** of values, which might just be a set of 1 value (known as a singleton set), a set of no values (i.e. the empty set $\varnothing$), or an infinite set (in the last example above, this was $\mathbb{R}$)

✎ The solutions to an equation are known as the **solution set**.

Algebra and Functions

Chapter 3 · Original slide 4

Solutions sets

4

For simultaneous equations, the same is true, except each ‘solution’ in the solution set is an assignment to **multiple** variables.
All equations have to be satisfied **at the same time, i.e. ‘simultaneously’**.

Scenario	Example	Solution Set
A single solution:		
Two solutions:		
No solutions:		
Infinitely large set of solutions:		

Chapter 3 · Original slide 5

1 :: Simultaneous Equations

5

Recap!

Solve the simultaneous equations:

$$3x + y = 8$$
$$2x - 3y = 9$$

We can either use substitution (i.e. making x or y the subject of one equation, and substituting it into the other) or elimination, but the latter is easier for linear equations.

$$\mathbf{9x+3y=24}$$

Solve the simultaneous equations:

$$x + 2y = 3$$
$$x^2 + 3xy = 10$$

We can’t use elimination this time as nothing would cancel.
We instead:
(1) Rearrange linear equation to make x or y the subject.
(2) Substitute into quadratic equation and solve.

Algebra and Functions

Chapter 3 · Original slide 6

Test Your Understanding

6

Solve the simultaneous equations:
$3x^2 + y^2 = 21$
$y = x + 1$

Chapter 3 · Original slide 7

Exercise 3A/B – Y1 Pages 40, 41

7

4 Solve the simultaneous equations:
$x + y = 3, \quad x^2 - 3y = 1$

Algebra and Functions

Chapter 3 · Original slide 8

Exercise 3A/B – Y1 Pages 40, 41

8

5 a By eliminating y from the equations $y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, show that $x^2 - 2x - 11 = 0$. **b** Hence, or otherwise, solve the simultaneous equations $y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, giving your answers in the form $a \pm b\sqrt{3}$, where a, b are integers.

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Exercise 3A/B – Y1 Pages 40, 41

9

6 One pair of solutions for the simultaneous equations $y = kx - 5, \quad 4x^2 - xy = 6$ is $(1, p)$ where k, p ar

a Find the values of k, p.

b Find the second pair of solutions for the simultaneous equations.

Algebra and Functions

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Extension

10

1 *[MAT 2012 1G]* There are *positive* real numbers x and y which solve the equations
$2x + ky = 4$,
$x + y = k$
for:
A) All values of k;
B) No values of k;
C) $k = 2$ only;
D) Only $k > -2$

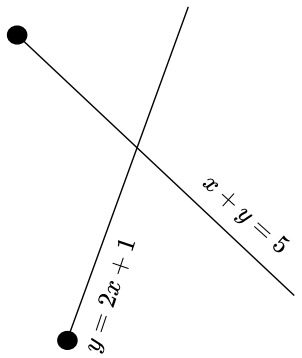
If $\mathbf{k=2}$ then $\mathbf{2x+2y=4}$ and $\mathbf{x+y=2}$ which are equivalent.

2 *[STEP 2010 Q1]* Given that
$5x^2 + 2y^2 - 6xy + 4x - 4y \equiv a(x - y + 2)^2 + b(cx + y)^2 + d$
a) Find the values of a, b, c, d.
b) Solve the simultaneous equations:
$5x^2 + 2y^2 - 6xy + 4x - 4y = 9$,
$6x^2 + 3y^2 - 8xy + 8x - 8y = 14$
(Hint: Can we use the same method in (a) to rewrite the second equation?)

Chapter 3 · Original slide 11

Simultaneous Equations and Graphs

11



Recall that a line with a given equation is **the set of all points that satisfy the equation**.

i.e. It is a graphical representation of the solution set where each (x, y) point represents each of the solutions x and y to the equation.
e.g. $x = 3, y = 7$

Now suppose we introduced a second simultaneous equation:
$y = 2x + 1$
$x + y = 5$

The second line again consists of all points (x, y) which satisfy the equation. So what point must satisfy both equations simultaneously?
The point of intersection!

Algebra and Functions

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Example

12

a) On the same axes, draw the graphs of:
$2x + y = 3$
$y = x^2 - 3x + 1$

b) Use your graph to write down the solutions to the simultaneous equations.

c) What algebraic method (perhaps thinking about the previous chapter), could we have used to show the graphs would have intersected twice?

Substituting linear equation into quadratic:
$\mathbf{y = 3 - 2x}$

Chapter 3 · Original slide 13

Another Example

13

a) On the same axes, draw the graphs of:
$y = 2x - 2$
$y = x^2 + 4x + 1$

b) Prove algebraically that the lines never meet.

Algebra and Functions

Chapter 3 · Original slide 14

Final Example

14

The line with equation $y = 2x + 1$ meets the curve with equation $kx^2 + 2y + (k - 2) = 0$ at exactly one point. Given that k is a positive constant:
a) Find the value of k.
b) For this value of k, find the coordinates of this point of intersection.

a

b

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Exercise 3C – Y1 Page 45

15

8 Given the simultaneous equations $2x - y = 1$, $x^2 + 4ky + 5k = 0$, where k is a non-zero constant,

a show that $x^2 + 8kx + k = 0$.
b Given that $x^2 + 8kx + k = 0$ has equal roots, find the value of k.
c for this value of k, find the solution of the simultaneous equations.

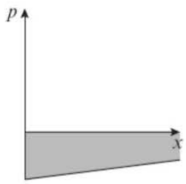
Algebra and Functions

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Exercise 3C – Y1 Page 45

16

9 A swimmer dives into a pool. Her position, p m, underwater can be modelled in relation to her horizontal distance, x m, from the point she entered the water as a quadratic equation $p = \frac{1}{2}x^2 - 3x$. The position of the bottom of the pool can be modelled by the linear equation $p = 0.3x - 6$. Determine whether this model predicts that the swimmer will touch the bottom of the pool.



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Set Builder Notation

17

Recall that a **set** is a **collection of values** such that:
a) The **order of values does not matter**.
b) There are **no duplicates**.

Recap from GCSE:

- We use curly braces to list the values in a set, e.g. $A = \{1, 4, 6, 7\}$
- If A and B are sets then $A \cap B$ is the **intersection** of A and B, giving a set which has the elements in A **and** B.
- $A \cup B$ is the **union** of A and B, giving a set which has the elements in A **or** in B.
- $\varnothing$ is the empty set, i.e. the set with nothing in it.
- Sets can also be infinitely large. $\mathbb{N}$ is the set of natural numbers (all positive integers), $\mathbb{Z}$ is the set of all integers (including negative numbers and 0) and $\mathbb{R}$ is the set of all real numbers (including all possible decimals).
- We write $x \in A$ to mean "x is a member of the set A". So $x \in \mathbb{R}$ would mean "x is a real number".
-

$$\{1, 2, 3\} \cap \{3, 4, 5\} =$$
$$\{1, 2, 3\} \cup \{3, 4, 5\} =$$
$$\{1, 2\} \cap \{3, 4\} =$$

Algebra and Functions

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Set Builder Notation

18

It is possible to construct sets without having to explicitly list its values. We use:

The | or : means "such that".

$$\{expr \mid condition\}$$
or
$$\{expr : condition\}$$

Can you guess what sets the following give?

$\{2x : x \in$ ____

$\{2^x : x \in$ ____

$\{xy : x, yareprime$ ____

Chapter 3 · Original slide 19

Set Builder Notation

19

Can you use set builder notation to specify the following sets?

All odd numbers. ____

All (real) numbers greater than 5. ____

All (real) numbers less than 5 **or** greater than 7. ____

All (real) numbers between 5 and 7 inclusive. ____

Algebra and Functions

Chapter 3 · Original slide 20

Recap of linear inequalities

20

Inequality	Solution Set
$2x + 1 > 5$	
$3\,(x - 5) \geq 5 - 2(x - 8)$	
$-x \geq 2$	

Combining Inequalities:

If $x < 3$ and $2 \leq x < 4$, what is the combined solution set?

Chapter 3 · Original slide 21

RECAP :: Solving Quadratic Inequalities

21

Solve $x^2 + 2x - 15 > 0$

Step 1: Get 0 on one side

Step 2: Factorise

Step 3: Sketch and reason

Algebra and Functions

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Solving Quadratic Inequalities

22

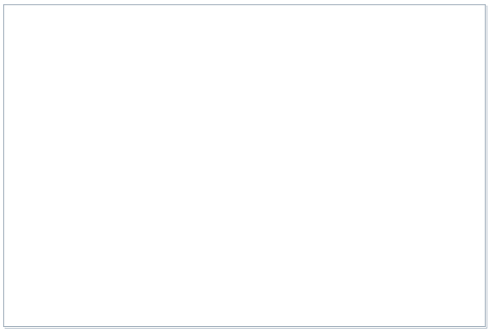
Solve $x^2 + 2x - 15 \leq 0$

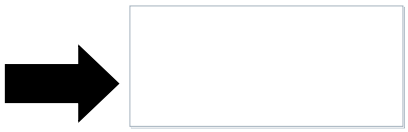
Step 1: Get 0 on one side

$(x + 5)(x - 3) \leq 0$

Step 2: Factorise

Step 3: Sketch and reason





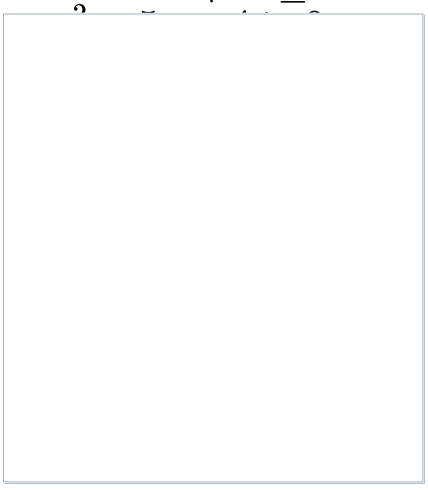
Chapter 3 · Original slide 23

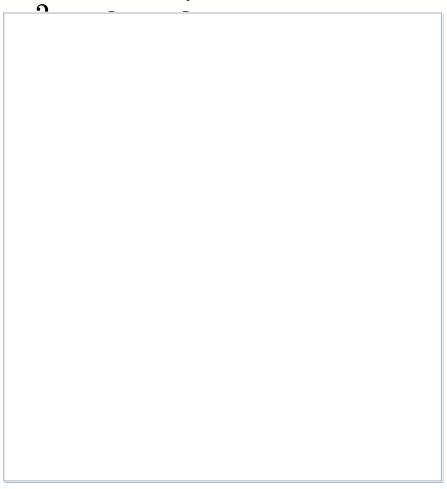
Further Examples

23

Solve $x^2 + 5x \geq -4$

Solve $x^2 < 9$





Note: do not skip the 'sketch step'.

To use the quadratic inequality solver on ClassWiz, go to
Menu → Inequalities, then choose 'order 2' (i.e. quadratic)
Try the same on a CG50.

Algebra and Functions

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Test Your Understanding

24

Edexcel C1 May 2010 Q3

Find the set of values of x for which (a) $3(x-2) < 8-2x$, (2) (b) $(2x-7)(1+x) < 0$, (3) (c) both $3(x-2) < 8-2x$ and $(2x-7)(1+x) < 0$. (1)

(a)

(b)

(c)

Edexcel C1 June 2008 Q8

Given that the equation $2qx^2 + qx - 1 = 0$, where q is a constant, has no real roots, (a) show that $q^2 + 8q < 0$. (2) (b) Hence find the set of possible values of q. (3)

(a)

(b)

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Deal with inequalities with a division by x

25

Find the set of values for which $\frac{6}{x} > 2, \quad x \neq 0$

Why can't we just multiply both sides by x?

Algebra and Functions

Chapter 3 · Original slide 26

Exercise 3D/3E – Y1 Page 47-48, 50-51

26

3 Use set notation to describe the set of values of x for which:

g $4(3x+7) \leq 20$ or $2(3x-5) \geq \frac{7-6x}{2}$

6 Find the set of values of x for which $x^2 - 5x - 14 > 0$.

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Exercise 3D/3E – Y1 Page 47-48, 50-51

27

8 Given that $x \neq 3$, find the set of values for which $\frac{5}{x-3} < 2$.

Algebra and Functions

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Exercise 3D/3E – Y1 Page 47-48, 50-51

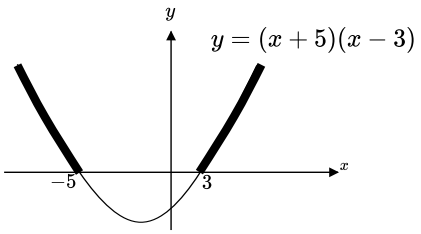
28

9 The equation $kx^2 - 2kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < 3$.

Chapter 3 · Original slide 29

Inequalities on Graphs

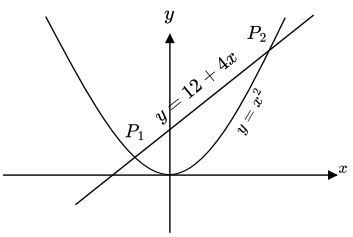
29



When we solved quadratic inequalities, e.g. $(x + 5)(x - 3) > 0$
We plotted $y = (x + 5)(x - 3)$ and observed the values of x for which $y > 0$.

Can we use a similar method when we don't have 0 on one side?

Example: L_1 has equation $y = 12 + 4x$. L_2 has equation $y = x^2$.
The diagram shows a sketch of L_1 and L_2 on the same axes.
a) Find the coordinates of P_1 and P_2, the points of intersection.
b) Hence write down the solution to the inequality $12 + 4x > x^2$.



Algebra and Functions

Chapter 3 · Original slide 30

Inequality Regions

30

On graph paper, shade the region that satisfies the inequalities:
$2y + x < 14$
$y \geq x^2 - 3x - 4$

You did this at GCSE, the only difference here being that the graphs involved might not be straight lines.

Step 2:
An inequality involving x and y represents a 2D region in space. Identify the correct side of each line each inequality represents.

Click to sketch >

Tip: To quickly sketch $2y + x = 14$, consider what happens when x is 0 and when y is 0.

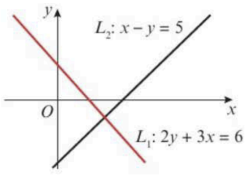
Tip: Make sure y is on the side where it is positive.
If y is on the smaller side, you're below the line.
If y is on the greater side, you're above the line.

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Exercise 3F/3G – Y1 Page 53, 55

31

1 L_1 has equation $2y + 3x = 6$. L_2 has the equation $x - y = 5$. The diagram shows a sketch of L_1, L_2.
a Find the coordinates of P, the point of intersection.
b Hence write down the solution to the inequality $2y + 3x > x - y$.



Algebra and Functions

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Exercise 3F/3G – Y1 Page 53, 55

32

3 Find the set of values of x for which the curve with equation $y = f(x)$ is below the line with equation $y = g(x)$.

a $f(x) = 3x^2 - 2x - 1$, $g(x) = x + 5$

b $f(x) = 2x^2 - 4x + 1$, $g(x) = 3x - 2$

c $f(x) = 5x - 2x^2 - 4$, $g(x) = -2x - 1$



Chapter 4 · Original slide 21

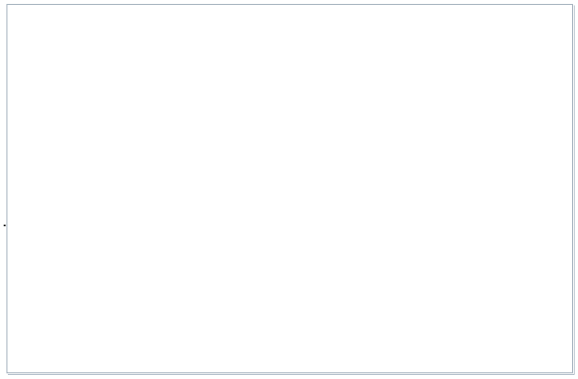
GCSE RECAP: Reciprocal Graphs

21

Sketch $y = \frac{1}{x}$



Sketch $y = -\frac{3}{x}$





Algebra and Functions

Chapter 4 · Original slide 22

Reciprocal Graphs

22

Sketch $y = \frac{3}{x^2}$

This is new to the A Level 2017 syllabus.

Sketch $y = -\frac{4}{x^2}$

Hint: Note that anything squared will always be at least 0.

Chapter 4 · Original slide 23

Reciprocal Graphs

23

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

Algebra and Functions

Chapter 4 · Original slide 24

Exercise 4C – Pure Y1 page 67

24

1 Use a separate diagram to sketch each pair of graphs. **a** $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$ **d** $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

Chapter 4 · Original slide 25

Exercise 4C – Pure Y1 page 67

25

2 Use a separate diagram to sketch each pair of graphs. **a** $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$

Algebra and Functions

Chapter 5 · Original slide 43

What's the point of straight line equations?

We saw in Chapter 2 that lots of things in real life have a 'quadratic' relationship, e.g. vertical height with time. Lots of real life variables have a 'linear' relationship, i.e. **there is a fixed increase/decrease in one variable each time the other variable goes up by 1 unit.**

Examples

Car sales made and take home pay.



Temperature and altitude (in a particular location)



The relationship between Celsius and Fahrenheit.

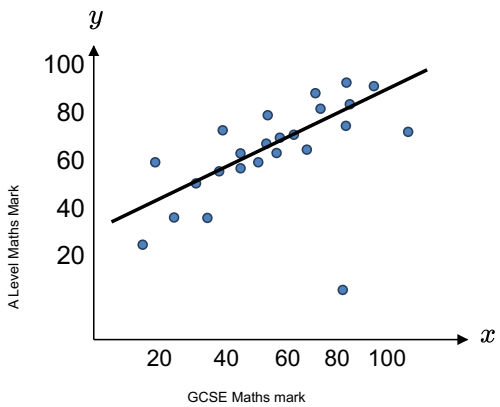


(And a pure maths one:)
The nth term of an arithmetic series.

3, 5, 8, 11, 14, …

Chapter 5 · Original slide 44

Modelling



A mathematical model is an attempt to model a real-life situation based on mathematical concepts.

For this example, **our model might be a linear model with equation $\mathbf{y=mx+c}$** where x is a student's GCSE mark and y is the predicted A Level mark.

Such a linear model can be drawn as a line of best fit.
The data obviously doesn't fit this line exactly. This chosen model may only partially fit the data (and the further the points are away from the line, the less suitable this model is).
We might decide another model, e.g. $y = ax^2 + bx + c$, is more appropriate.

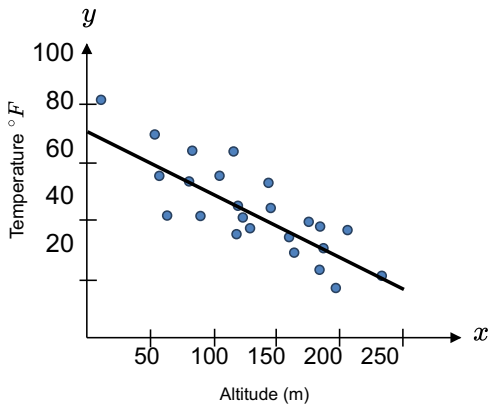
But if we choose a well-known model such as a linear one, then we can use **established mathematical theory** in useful ways:

- We need to choose the most appropriate 'parameters' m and c so the model best matches the data. You will learn in S1 there are existing techniques to do this.
- We can then predict a student's A Level mark based on their maths mark.

Algebra and Functions

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Example



The temperature y at different points on a mountain is recorded at different altitudes x.
Suppose we were to use a linear model $y = mx + c$.

a Determine m and c (you can assume the line goes through $(0, 70)$ and $(250, 20)$.

Interpret the meaning of m

b **The temperature at sea l** **n $0.2°\mathbf{F}$ for each extra**

c

Chapter 5 · Original slide 46

Evaluating if a linear model is most suitable

Choosing a model for our data usually involves making **simplifying assumptions**. For a linear model we are assuming:

The y value goes up the same amount for each unit increase in x.

The current population of Bickerstonia is 26000. This year (2017) the population increased by 150. Matt decides to model the population P based on the years t after 2017 by the linear model:

$$P = mt + c$$

Why might this not be a suitable model?

Algebra and Functions

Chapter 5 · Original slide 47

Exercise 5H – Pure Y1 Page 106-108

3 The cost of electricity, E, in pounds and the number of kilowatt hours, h, are shown in the table.

kilowatt hours, h	0	15	40	60	80	110
cost of electricity, E	45	46.8	49.8	52.2	54.6	58.2

a Draw a graph of the data.
b Explain how you know a linear model would be appropriate.
c Deduce an equation in the form $E = ah + b$.
d Interpret the meaning of the coefficients a and b.
e Use the model to find the cost of 65 kilowatt hours.

Chapter 5 · Original slide 48

Exercise 5H – Pure Y1 Page 106-108

6 The average August temperature in Exeter is 20°C or 68°F. The average January temperature in the same place is 9°C or 48.2°F.
a Write an equation linking Fahrenheit F and Celsius C in the form $F = aC + b$. \hfill(3 marks)
b Interpret the values of a and b. \hfill(2 marks)
c The highest temperature recorded in the UK was 101.3°F. Calculate this temperature in Celsius.

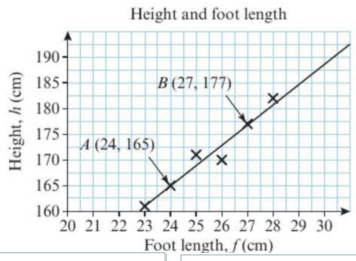
d For what value is the temperature in Fahrenheit the same as the temperature in Celsius?

Algebra and Functions

Chapter 5 · Original slide 49

Exercise 5H – Pure Y1 Page 106-108

8 The scatter graph shows the height h and foot length f of 8 students. A line of best fit is drawn on the scatter graph.
a Explain why the data can be approximated to a linear model. \hfill(1 mark)
b Use points A and B on the scatter graph to write a linear equation in the form $h = af + b$. \hfill(3 marks)
c Calculate the expected height of a person with a foot le
26.5 cm. \hfill(1 mark)



9 The price P of a good and the quantity Q of a good are linked.
The demand for a new pair of trainers can be modelled using the equation $P = -\frac{3}{4}Q + 35$.
The supply of the trainers can be modelled using the equation $P = \frac{2}{3}Q + 1$.
a Draw a sketch showing the demand and supply lines on the same pair of axes.
The equilibrium point is the point where the supply and demand lines meet.
b Find the values of P and Q at the equilibrium point.

Chapter 7 · Original slide 4

1: Simplifying Algebraic Fractions

4

Recall that you can simplify fractions by **dividing** the numerator and denominator by a **common factor**.

Hint: To identify common factors we need to factorise first.

$$\frac{x^2-1}{x^2+x}$$

$$\frac{x^2+3x+2}{x+1} =$$

$$\frac{2x^2+11x+12}{x^2+9x+20} =$$

$$\frac{4-x^2}{x^2+2x-8} =$$

Algebra and Functions

Chapter 7 · Original slide 5

Exercise 7A – Y1 Pages 138-139

5

2 *Simplify these fractions as far as possible :*

m $\frac{2x^2+3x+1}{x^2-x-2}$ **n** $\frac{x^2+6x+8}{3x^2+7x+2}$ **o** $\frac{2x^2-5x-3}{2x^2-9x+9}$

3 $\frac{6x^3+3x^2-84x}{6x^2-33x+42} = \frac{ax(x+b)}{x+c}$, where a, b and c are constants.
Work out the values of a, b and c.

Chapter 7 · Original slide 6

Terminology

6

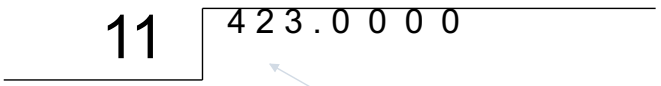
$$11 \div 4 = 2rem3$$

Algebra and Functions

Chapter 7 · Original slide 7

Normal Long Division

7



1. We found how many whole number of times (i.e. the quotient) the divisor went into the dividend.

2. We multiplied the quotient by the dividend.

3. …in order to find the remainder.

4. Find we 'brought down' the next number.

Chapter 7 · Original slide 8

Example

8

How many times does x go into $6x^3$?

$$6x^2 - 2x + 3$$

Now repeat! How many times does x go into $-2x^2$?

$$x + 5 \;\overline{)\; 6x^3 + 28x^2 - 7x + 15}$$

Multiply $6x^2$ by $(x + 5)$. The first term should match with above.

$$6x^3 + 30x^2$$

Subtract and carry down next term.

$$-2x^2 - 7x$$

$$-2x^2 - 10x$$

$$3x + 15$$

$$3x + 15$$

This is the remainder.

$$0$$

Tip:
You can check your solution by expanding:
$(x + 5)(6x^2 - 2x + 3)$
But if you know you should get no remainder, ending with 0 at the bottom is a good sign!

Tip:
Be **very** careful subtracting negatives.
$-7x - (-10x) = 3x$

Algebra and Functions

Chapter 7 · Original slide 9

Further Example

9

Find the remainder when $3x^3 - 2x + 4$ is divided by $x - 1$.

x - 1 | $3x^3 + 0x^2 - 2x + 4$

Note: Ensure that any missing terms in the polynomial are filled in, so that the powers decrease by 1 each time.

Chapter 7 · Original slide 10

Test Your Understanding

10

Find the remainder when $2x^3 - 5x^2 - 16x + 10$ is divided by $x - 4$.

Algebra and Functions

Chapter 7 · Original slide 11

Further Test Your Understanding

11

Divide $8x^3 - 1$ by $2x - 1$.

Chapter 7 · Original slide 12

Exercise 7B – Y1 Pages 142

12

15 $\mathrm{f}(x) = 10x^3 + 43x^2 - 2x - 10$.
Find the remainder when $\mathrm{f}(x)$ is divided by $(5x + 4)$.

16 $\mathrm{f}(x) = 3x^3 - 14x^2 - 47x - 14$.
a Find the remainder when $\mathrm{f}(x)$ is divided by $(x - 3)$.
b Given that $(x + 2)$ is a factor of $\mathrm{f}(x)$, factorise $\mathrm{f}(x)$ completely.

Algebra and Functions

Chapter 7 · Original slide 13

Exercise 7B – Y1 Pages 142

13

17 a Find the remainder when $x^3 + 6x^2 + 5x - 12$ is divided by
i $x - 2$,
ii $x + 3$.
b Hence, or otherwise, find all the solutions to the equation $x^3 + 6x^2 + 5x - 12 = 0$.

Chapter 7 · Original slide 14

Exercise 7B – Y1 Pages 142

14

18 $\mathrm{f}(x) = 2x^3 + 3x^2 - 8x + 3$.
a Show that $\mathrm{f}(x) = (2x - 1)(ax^2 + bx + c)$ where a, b and c are constants to be found.
b Hence factorise $\mathrm{f}(x)$ completely.
c Write down all the real roots of the equation $\mathrm{f}(x) = 0$.

Algebra and Functions

Chapter 7 · Original slide 15

Exercise 7B – Y1 Pages 142

15

19 $\mathrm{f}(x) = 12x^3 + 5x^2 + 2x - 1$.
a Show that $(4x - 1)$ is a factor of $\mathrm{f}(x)$ and write $\mathrm{f}(x)$ in the form $(4x - 1)(ax^2 + bx + c)$.
b Hence, show that the equation $12x^3 + 5x^2 + 2x - 1 = 0$ has exactly 1 real solution.

Chapter 7 · Original slide 16

3: The Factor Theorem

16

$$x^3 + x^2 - 4x - 4 = (x - 2)\left(x^2 + 3x + 2\right)$$

We can see that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.
What would happen if x is 2?

Algebra and Functions

Chapter 7 · Original slide 17

Examples (in textbook)

17

Show that $(x-2)$ is a factor of x^3+x^2-4x-4.

Fully factorise $2x^3+x^2-18x-9$.

Chapter 7 · Original slide 18

Using Factor Theorem to find unknown coefficients

18

Given that $2x+1$ is a factor of $6x^3+ax^2+1$,
determine the value of a.

Algebra and Functions

Chapter 7 · Original slide 19

Test Your Understanding

19

Edexcel C2 May 2016 Q2

$f(x) = 6x^3 + 13x^2 - 4$

No long in spec.

(a) Use the remainder theorem to find the remainder when $f(x)$ is divided by $(2x + 3)$. \hfill(2)
(b) Use the factor theorem to show that $(x + 2)$ is a factor of $f(x)$. \hfill(2)
(c) Factorise $f(x)$ completely. \hfill(4)

Given that $3x - 1$ is a factor of $3x^3 + 11x^2 + ax + 1$, determine the value of a.

$\mathbf{f}\left(\frac{1}{3}\right.$
$\frac{7}{3}+$

Chapter 7 · Original slide 20

Exercise 7C – Y1 Pages 146

20

13 $f(x) = 3x^3 - 12x^2 + 6x - 24$.
a Use the factor theorem to show that $(x - 4)$ is a factor of $f(x)$.
b Hence, show that 4 is the only real root of the equation $f(x) = 0$.

Algebra and Functions

Chapter 7 · Original slide 21

Exercise 7C – Y1 Pages 145-146

21

14 $\mathrm{f}(x) = 4x^3 + 4x^2 - 11x - 6$.
a Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$.
b Factorise $\mathrm{f}(x)$ completely.

Chapter 7 · Original slide 22

Exercise 7C – Y1 Pages 145-146

22

15 a Show that $(x - 2)$ is a factor of $9x^4 - 18x^3 - x^2 + 2x$.
b Hence, find four real solutions to the equation $9x^4 - 18x^3 - x^2 + 2x = 0$.

Algebra and Functions

Chapter 7 · Original slide 23

Exercise Extension

23

1 *[MAT 2006 1E]* The cubic
$$x^3 + ax + b$$
Has both $(x-1)$ and $(x-2)$ as factors. Determine the values of a and b.

$\mathbf{a=-7, b=6}$

[MAT 2009 1I]

2 $$n^2x^{2n+3} - 25nx^{n+1} + 150x^7$$
Has $x^2 - 1$ as a factor
A) for no values of n;
B) for $n = 10$ only;
C) for $n = 15$ only;
D) for $n = 10$ and $n = 15$ only.

$\mathbf{x^2-1 = (x+1)(x-1)}$ **so both factors.**
$$\mathbf{f(1) = n^2 - 25n + 150 = 0}$$
If n is odd: $\mathbf{f(-1) = -n^2 - 25n - 150 = 0}$

The **remainder theorem** states that if $f(x)$ is divided by $(x-a)$, the remainder is $f(a)$. This similarly works whenever a makes the divisor 0.

3 [MAT 2013 1G] Let $n \geq 2$ be an integer and $p_n(x)$ be the polynomial
$$p_n(x) = (x-1) + (x-2) + \ldots + (x-n)$$
What is the remainder, in terms of n, when $p_n(x)$ is divided by $p_{n-1}(x)$?

Chapter 8 · Original slide 4

Starter

4

a) Expand $(a+b)^0$
b) Expand $(a+b)^1$
c) Expand $(a+b)^2$
d) Expand $(a+b)^3$
e) Expand $(a+b)^4$

What do you notice about:

The coefficients:
The powers of a and b:

4

Algebra and Functions

Chapter 8 · Original slide 5

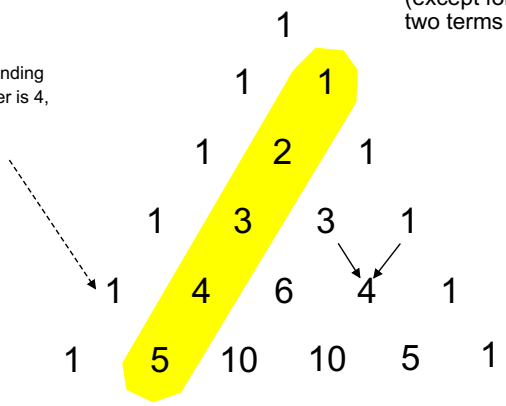
More on Pascal's Triangle

5

The second number of each row tells us what row we should use for an expansion.

So if we were expanding $(2+x)^4$, the power is 4, so we use this row.

In Pascal's Triangle, each term (except for the 1s) is the sum of the two terms above.



Tip: memorise each row up to what you see here.

We'll see later WHY each row gives us the coefficients in an expansion of $(a+b)^n$

5

Chapter 8 · Original slide 6

Example

6

Find the expansion of $(2+3x)^4$

$$(2+3x)^4 =$$

Next have descending or ascending powers of one of the terms, going between 0 and 4 (note that if the power is 0, the term is 1, so we need not write it).

First fill in the correct row of Pascal's triangle.

Simplify each term (ensuring any number in a bracket is raised to the appropriate power)

And do the same with the second term but with powers going the opposite way, noting again that the 'power of 0' term does not appear.

Tip: Initially write *one line per term* for your expansion (before you simplify at the end), as we have done above. There will be less faffing trying to ensure you have enough space for each term.

6

Algebra and Functions

Chapter 8 · Original slide 7

Another Example

7

$(1-2x)$ is the same as $(1+(-2x))$, so we expand as before, but use $-2x$ for the second term.

$$(1-2x)^3 =$$

Tip: If one of the terms in the original bracket is negative, the terms in your expansion will oscillate between positive and negative. If they don't (e.g. two consecutive negatives), you've done something wrong!

7

Chapter 8 · Original slide 8

Getting a single term in the expansion

8

The coefficient of x^2 in the expansion of $(2-cx)^5$ is 720.
Find the possible value(s) of the constant c.

$\vdots$

8

Algebra and Functions

Chapter 8 · Original slide 9

Test Your Understanding

9

Edexcel C2

(a) Find the first 3 terms, in ascending powers of x, of the binomial expansion of
$(2 + kx)^7$
where k is a constant. Give each term in its simplest form. \hfill(4)
Given that the coefficient of x^2 is 6 times the coefficient of x,
(b) find the value of k. \hfill(2)

(a) $(7$

$(2 + k$

\hfill B
(If 672
(b) $6 \times$
$k = 4$
\hfill[6

9

Chapter 8 · Original slide 10

Exercise 8A – Y1 page 160-161

10

4 Fully expand the expression $(1 + 3x)(1 + 2x)^3$.

5 Expand $(2 + y)^3$. Hence or otherwise, write down the expansion of $(2 + x - x^2)^3$ in ascending powers of x.

10

Algebra and Functions

Chapter 8 · Original slide 11

Exercise 8A – Y1 page 160-161

11

6 The coefficient of x^2 in the expansion of $(2+ax)^3$ is 54. Find the possible values of the constant a.

7 The coefficient of x^3 in the expansion of $(2-x)(3+bx)^3$ is 45. Find possible values of the constant b.

11

Chapter 8 · Original slide 12

Exercise 8A – Y1 page 160-161

12

8 Work out the coefficient of x^2 in the expansion of $(p-2x)^3$. Give your answer in terms of p.

9 After 5 years, the value of an investment of \pounds500 at an interest rate of $X\%$ per annum is given by $500\left(1+\frac{X}{100}\right)^5$

Find an approximation for this expression in the form $A+BX+CX^2$, where A, B and C are constants to be found. You can ignore higher powers of X.

12

Algebra and Functions

Chapter 8 · Original slide 13

Extension

13

[MAT 2009 1J]

The number of pairs of positive integers x, y which solve the equation:

$$x^3 + 6x^2y + 12xy^2 + 8y^3 = 2^{30}$$

is:

A) 0
B) 2^6
C) $2^9 - 1$
D) $2^{10} + 2$
E)

The LHS is the binomial expansion of $(x+2y)^3$, therefore:

13

Chapter 8 · Original slide 14

Factorial and Choose Function

14

$n! = n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1$
said "n factorial", is the number of ways of arranging n objects in a line.

For example, suppose you had three letters, A, B and C, and wanted to arrange them in a line to form a 'word', e.g. ACB or BAC.

- There are 3 choices for the first letter.
- There are then 2 choices left for the second letter.
- There is then only 1 choice left for the last letter.
-

There are therefore $3 \times 2 \times 1 = 3! = 6$ possible combinations.

Your calculator can calculate a factorial using the $x!$ button.

$$nCr = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

said "n choose r", is the number of ways of 'choosing' r things from n, such that the order in our selection does not matter.
These are also known as **binomial coefficients**.

For example, if you a football team captain and need to choose 4 people from amongst 10 in your class, there are $\binom{10}{4} = \frac{10!}{4!6!} = 210$ possible selections.

(Note: the $\binom{10}{4}$ notation is preferable to $10C4$)

Use the nCr button on your calculator (your calculator input should display "10C4")

14

Algebra and Functions

Chapter 8 · Original slide 15

Examples

15

Calculate the value of the following. You may use the factorial button, but not the nCr button.

a) $5!$

b) $\binom{5}{3}$

c) $0!$

d) $\binom{20}{1}$

e) $\binom{20}{0}$

f) $\binom{20}{2}$

g) $\binom{20}{18}$

a $1 = 120$

b $= 10$

c

d

e

f

g

15

Chapter 8 · Original slide 16

Why do we care?

16

If the power in the binomial expansion is large, e.g. $(x+3)^{20}$, it is no longer practical to go this far down Pascal's triangle. We can instead use the choose function to get numbers from anywhere within the triangle. We'll practise doing this after the next exercise.

1

Notice: The top number matches the row number. The bottom number goes from 0 and eventually matches the top number. It's easy to see from the symmetry of Pascal's Triangle that $\binom{4}{1} = \binom{4}{3}$ for example.

1 3 3 1

1 4 6 4 1

0th row $\binom{0}{0}$

1st row $\binom{1}{0}$ $\binom{1}{1}$

2nd row $\binom{2}{0}$ $\binom{2}{1}$ $\binom{2}{2}$

3rd row $\binom{3}{0}$ $\binom{3}{1}$ $\binom{3}{2}$ $\binom{3}{3}$

$\binom{4}{0}$ $\binom{4}{1}$ $\binom{4}{2}$ $\binom{4}{3}$ $\binom{4}{4}$

Textbook Note: The textbook refers to the top row as the "1st row" and the first number in each row as the "1st entry". This might sound sensible, but is against accepted practice: It makes much more sense that the row number matches the number at the top of the binomial coefficient, and the entry number matches the bottom number. We therefore call the top row the "0th row" and the first entry of each row the "0th entry". So the kth entry of the nth row of Pascal's Triangle is therefore a nice clean $\binom{n}{k}$, not $\binom{n-1}{k-1}$ as suggested by the textbook.

16

Algebra and Functions

Chapter 8 · Original slide 17

Interesting facts

17

(You are not required to know this, but it is helpful for STEP)

$$\binom{0}{0}$$

$$\binom{1}{0}\quad\binom{1}{1}$$

$$\binom{2}{0}\quad\binom{2}{1}\quad\binom{2}{2}$$

$$\binom{3}{0}\quad\binom{3}{1}\quad\binom{3}{2}\quad\binom{3}{3}$$

$$\binom{4}{0}\quad\binom{4}{1}\quad\binom{4}{2}\quad\binom{4}{3}\quad\binom{4}{4}$$

We earlier saw that each entry of Pascal's Triangle is the sum of the two above it. Thus for example:

$$\binom{3}{1}+\binom{3}{2}=\binom{4}{2}$$

More generally:

$$\binom{n-1}{k-1}+\binom{n-1}{k}=\binom{n}{k}$$

This is known as Pascal's Rule.

Informal proof of Pascal's Rule:

Suppose I have n items and I have to choose k of them. Clearly there's $\binom{n}{k}$ possible selections. But we could also find the number of selections by considering the first item of the n available:

- It might be chosen. If so, we have $k-1$ items left to choose from amongst the $n-1$ remaining. That's $\binom{n-1}{k-1}$ possible selections.
- Otherwise it is not chosen. We still have k items to choose, from amongst the remaining $n-1$ items. That's $\binom{n-1}{k}$ possible selections.

Thus in total there are $\binom{n-1}{k-1}+\binom{n-1}{k}$ possible selections.

17

Chapter 8 · Original slide 18

Exercise 8B – Y1 page 162

18

10 Given that $\binom{50}{13}=\dfrac{50!}{13!a!}$, write down the value of a.

11 Given that $\binom{35}{p}=\dfrac{35!}{p!18!}$, write down the value of p.

18

Algebra and Functions

Chapter 8 · Original slide 19

Exercise 8B – Y1 page 162

19

Challenge
a Work out ${}^{10}C_3$ and ${}^{10}C_7$.
b Work out ${}^{14}C_5$ and ${}^{14}C_9$.
c What do you notice about your answers to parts **a** and **b**?
d Prove that ${}^{n}C_r = {}^{n}C_{n-r}$.

19

Chapter 8 · Original slide 20

Using Binomial Coefficients to Expand

20

In the previous section we learnt about the 'choose' function and how this related to Pascal's Triangle.

1 5 10 10 5 1

$$\binom{5}{0}\ \binom{5}{1}\ \binom{5}{2}\ \binom{5}{3}\ \binom{5}{4}\ \binom{5}{5}$$

Why do rows of Pascal's Triangle give us the coefficients in a Binomial Expansion?

Consider: $(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b)$
Each term of the expansion involves picking one term from each bracket.
How many times will a^3b^2 appear in the expansion?
To get $\mathbf{a^3b^2}$ we must have chosen 3 a's from the 5 brackets (the rest b's).

$\binom{5}{3}$ $\binom{5}{3}$

20

Algebra and Functions

Chapter 8 · Original slide 21

Using Binomial Coefficients to Expand

21

$\mathbb{N}$ is the set of natural numbers, i.e. positive integers. This formula is only valid for positive integers n. In Year 2 you will see how to deal with fractional/negative n.

The binomial expansion, when $n \in \mathbb{N}$:

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n$$

Find the first 4 terms in the expansion of $(3x+1)^{10}$, in ascending powers of x.

21

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Test Your Understanding

22

Find the first 3 terms in the expansion of $\left(2-\frac{1}{3}x\right)^7$, in ascending powers of x.

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Algebra and Functions

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Exercise 8C – Y1 page 164

23

5 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2-x)^6$ and simplify each term.

6 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(3-2x)^5$ giving each term in its simplest form.

7 Find the binomial expansion of $\left(x+\frac{1}{x}\right)^5$ giving each term in its simplest form.

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Extension

24

1 [AEA 2013 Q1a] In the binomial expansion of $\left(1+\frac{12n}{5}x\right)^n$ the coefficients of x^2 and x^3 are equal and non-zero. Find the possible values of n.

Hint: Remember that $\binom{n}{2}=\frac{n(n-1)}{2!}$

Can you similarly simplify $\binom{n}{3}$ using $\binom{n}{r}=\frac{n!}{\quad\quad}$?

2 [STEP I 2010 Q5a] By considering the expansion of $(1+x)^n$, where n is a positive integer, or otherwise, show that:

$$\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\ldots+\binom{n}{n}=2^n$$

reflection: This means that the sum of each row in Pascal's Triangle gives successive powers of 2. Safe!

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Algebra and Functions

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Getting a single term in the expansion

25

In the expansion of $(a+b)^n$ the general term is given by $\binom{n}{r}a^{n-r}b^r$

Expression	Power of x in term wanted.	Term in expansion
$(a+x)^{10}$	3	
$(2x-1)^{75}$	50	
$(3-x)^{12}$	7	
$(3x+4)^{16}$	3	

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Getting a single term in the expansion

26

The coefficient of x^4 in the expansion of $(1+qx)^{10}$ is 3360.
Find the possible value(s) of the constant q.

Term is:

•

Therefore:
$\mathbf{210q^4=3360}$
$\mathbf{q^4=16}$
$\mathbf{q=\pm 2}$

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Algebra and Functions

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Test Your Understanding

27

In the expansion of $(1+ax)^{10}$, where a is a non-zero constant the coefficient of x^3 is double the coefficient of x^2. Find the value of a.

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Exercise 8D – Y1 page 166-167

28

5 When $(1-2x)^p$ is expanded, the coefficient of x^2 is 40. Given that $p>0$, use this information to find:

a the value of the constant p
b the coefficient of x
c the coefficient of x^3

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Algebra and Functions

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Exercise 8D – Y1 page 166-167

29

6 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(5 + px)^{30}$, where p is a non-zero constant.
b Given that in this expansion the coefficient of x^2 is 29 times the coefficient of x work out the value of p.

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Exercise 8D – Y1 page 166-167

30

7 a Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 + qx)^{10}$, where q is a non-zero constant.
b Given that in the expansion of $(1 + qx)^{10}$ the coefficient of x^3 is 108 times the coefficient of x, work out the value of q.

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Algebra and Functions

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Exercise 8D – Y1 page 166-167

31

8 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(1+px)^{11}$, where p is a constant.
b The first 3 terms in the same expansion are $1, 77x$ and qx^2, where q is a constant. Find the value of p and the value of q.

$p = 7$

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Exercise 8D – Y1 page 166-167

32

9 a Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1+px)^{15}$, where p is a non-zero constant.
b Given that, in the expansion of $(1+px)^{15}$, the coefficient of x is $(-q)$ and the coefficient of x^2 is $5q$, find the value of p and the value of q.

$21p^2 = q$

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Algebra and Functions

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Exercise 8D – Y1 page 166-167

33

10 In the binomial expansion of $(1+x)^{30}$, the coefficients of x^9 and x^{10} are p and q respectively. Find the value of $\frac{q}{p}$.
\hfill(4 mar . . .

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Extension

34

1 *[MAT 2014 1G]* Let n be a positive integer. The coefficient of x^3y^5 in the expansion of $\left(1+xy+y^2\right)^n$ equals:

A) n
B) 2^n
C) $\binom{n}{3}\binom{n}{5}$
D) $4\binom{n}{4}$
E) $\binom{n}{8}$

Try to imagine n brackets written out. To get x^3y^5,

2 [STEP I 2013 Q6] By considering the coefficient of x^r in the series for $(1+x)(1+x)^n$, or otherwise, obtain the following relation between binomial coefficients:

$$\binom{n}{r}+\binom{n}{r-1}=\binom{n+1}{r}$$

Noting that $(1+x)(1+x)^n=(1+x)^{n+1}$,
x^r $\binom{n+1}{r}x^r$

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Algebraic proof

Using the definition of the choose function, you can show that $\binom{n}{r}+\binom{n}{r-1}=\binom{n+1}{r}$

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Estimating Powers

35

Edexcel C2 Jan 2012 Q3

(a) Find the first 4 terms of the binomial expansion, in ascending powers of x, of $\left(1+\frac{x}{4}\right)^8$, giving each term in its simplest form. \hfill(4)
(b) Use your expansion to estimate the value of $(1.025)^8$, giving your answer to 4 decimal places. \hfill(3)

Tip: Use your calculator to compare against the exact value of 1.025^8.

a

$\left(1\right.$

$=$

b

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Test Your Understanding

36

Edexcel C2 Jan 2008 Q3

(a) Find the first 4 terms of the expansion of $\left(1+\frac{x}{2}\right)^{10}$ in ascending powers of x, giving each term in its simplest form. \hfill(4)
(b) Use your expansion to estimate the value of $(1.005)^{10}$, giving your answer to 5 decimal places. \hfill(3)

(a)
$\left(1+\frac{1}{2}\right.$
\hfill M
$=1+$
[Allow
(b)
$\left(1+\frac{1}{2}\right.$

\hfill M

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Algebra and Functions

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Exercise 8E – Y1 Page 168-169

37

7 a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\left(3-\frac{x}{5}\right)^{10}$, giving each term in its simplest form. \hfill$(4\text{ m}\ldots$

b Explain how you would use your expansion to give an estimate for the value of 2.98^{10}. \hfill$(1\ldots$



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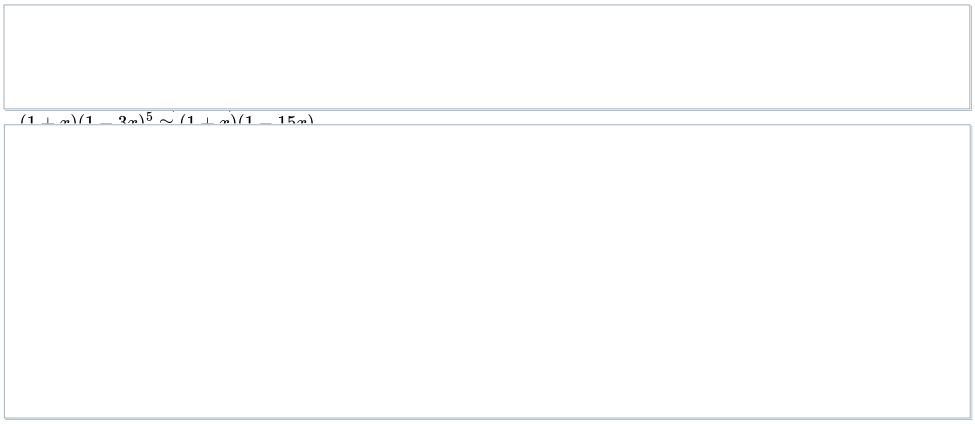
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Exercise 8E – Y1 Page 168-169

38

8 a Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(1-3x)^5$. Give each term in its simplest form. \hfill$(4\ldots$

b If x is small, so that x^2 and higher powers can be ignored, show that $(1+x)(1-3x)^5 \approx 1-14x$. \hfill$(2$



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Algebra and Functions

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Exercise 8E – Y1 Page 168-169

39

9 A microchip company models the probability of having no faulty chips on a single production run as:
$\mathrm{P}(\text{no fault}) = (1-p)^n,\ p < 0.001$
where p is the probability of a single chip being faulty, and n being the total number of chips produced.
a State why the model is restricted to small values of p. \hfill$(1\ \text{mar}\ldots$
b Given that $n = 200$, find an approximate expression for $\mathrm{P}(\text{no fault})$ in the form $a + bp + cp^2$. \hfill $(2\ \text{mark}\ldots$
c The company wants to achieve a 92% likelihood of having no faulty chips on a production run of 200 chips. Use your answer to part **b** to suggest a maximum value of p for this to be the case. \hfill$(4\ \text{m}$

$19900m^2 - 200m + 0.08 = 0$

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Chapter 10 · Original slide 29

Quadratics in sin/cos/tan

29

We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360°$.

Method 1: Use a substitution.

Method 2: Factorise without substitution.

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More Examples

30

Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360^\circ$.

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180^\circ \leq x \leq 180^\circ$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

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Test Your Understanding

31

Edexcel C2 Jan 2010 Q2

(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360^\circ$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

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Exercise 10F – Y1 P221-222

32

5 Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

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Exercise 10F – Y1 P221-222

33

7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

Algebra and Functions

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Exercise 10F – Y1 P221-222

34

9\quad $\tan^2 x - 2\tan x - 4 = 0$.
a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.
b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540°$.

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Quadratics in e^x

42

In previous chapters we've already dealt with quadratics in disguise, e.g. "quadratic in sin". We therefore just apply our usual approach: either make a suitable substitution so the equation **is** then quadratic, or (strongly recommended!) **go straight for the factorisation**.

Solve $e^{2x} + 2e^x - 15 = 0$

Solve $e^x - 2e^{-x} = 1$

Algebra and Functions

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Test Your Understanding

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Solve $\ln(3x+1) = 2$

Solve $e^{2x} + 5e^{x} = 6$

Solve $2^{x}e^{x+1} = 3$ giving your answer as an exact value.

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Exercise 14G – Y1 Page 327

44

4 *Find the exact solutions to the* equation $e^{x} + 12e^{-x} = 7$.

6 Solve $3^{x}e^{4x-1} = 5$, giving your answer in the form $\dfrac{a + \ln b}{c + \ln d}$